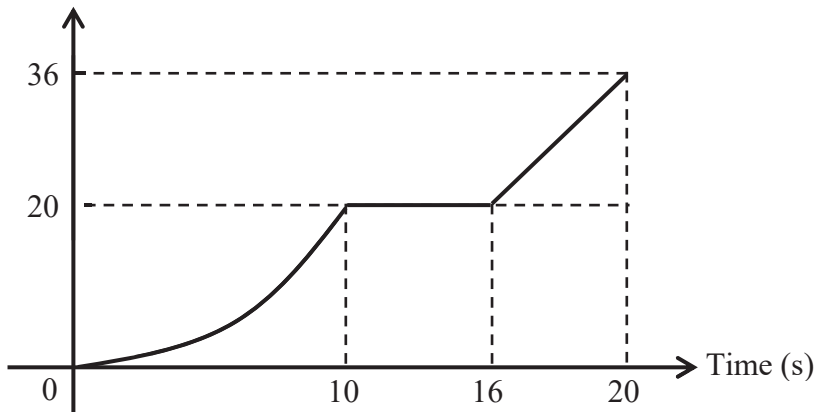


- 5 The diagram below shows the distance-time graph of a particle from point P for the first 20 seconds of its journey.

Distance from P (m)



- (a) Find the average speed during the first 10 seconds.

Answer m/s [1]

- (b) Find the acceleration of the particle when $t = 17$.

Answer m/s^2 [1]

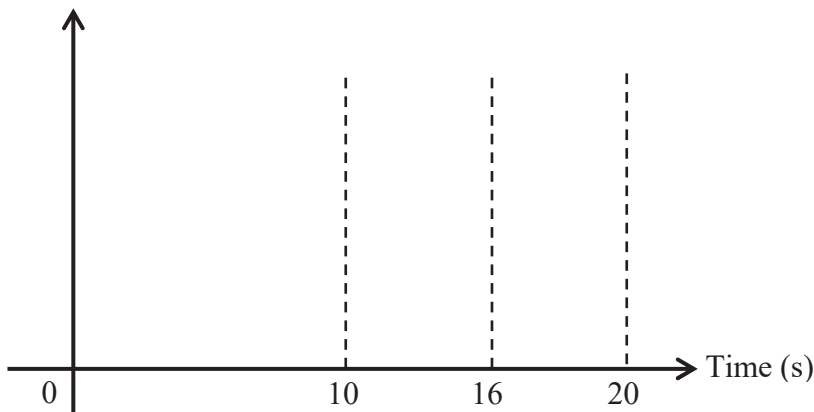
- (c) Find the speed of the particle when $t = 18$.

Answer m/s [1]

- (d) The speed increases uniformly for the first 10 seconds.
Sketch the speed-time graph for the first 20 seconds on the answer space,
indicating the corresponding speeds clearly.

[3]

Speed (m/s)

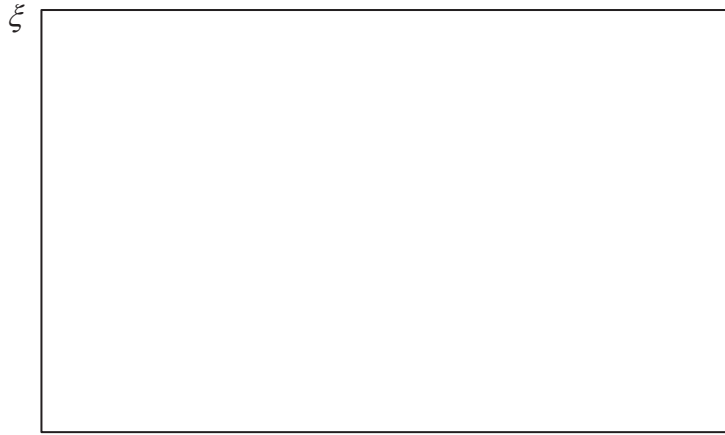


6 Given $\xi = \{x \text{ is an integer: } 10 \leq x < 22\}$, and

$$A = \{x : x \text{ is a prime number}\}$$

$$B = \{x : x \text{ is a odd number}\}$$

(a) Draw the Venn diagram to show the elements of ξ and two sets A and B . [1]



(b) List the elements in the set $A \cap B$.

Answer [1]

(c) Find $n(A \cup B')$.

Answer [1]

- 7 500 ml of liquid fertiliser was made by mixing 200 ml of water with a 300 ml mixture of chemical A and chemical B .
The ratio of chemical A to chemical B is 1 : 9.
Abigail wants to decrease the concentration of chemical B to 20% of the liquid fertiliser.
Calculate the volume of chemical B to be removed from the original 500 ml of liquid fertiliser.

Answer ml [2]
